Full Distance, Short Detour: A One-Query Forgery against FDSD

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This audit used the OpenAI model Daybreak Blue at the ultra reasoning level. OpenAI Codex generated and selected this paper's title and intentionally chose "Full Distance, Short Detour" as wordplay contrasting the scheme's full-distance decoding premise with the short public linear detour used by the forgery. OpenAI Codex performed AI-assisted target selection, reconstruction, cryptanalysis, implementation, experiments, and technical writing. A separate AI agent performed adversarial review on the retained key, and a fresh-context AI agent independently reproduced the attack on five fresh keys. The final exact-target history recheck and independent Publication Reviewer gate also passed. All required publication gates passed, and the result is classified as `VERIFIED_ATTACK`. No independent human verification was performed or required.

Abstract

We describe a public-only, one-signing-query fresh-message forgery against the Level-I parameters of Abdelhaliem Babiker's *A Signature Scheme from Full-Distance Syndrome Decoding*, IACR ePrint 2023/1956. The ordinary verifier accepts a weight-343 vector σ when

$$H^\star\sigma = Rh + R_{[25]}\zeta.$$

The public matrix $H^\star$ retains 125 hidden-support columns spanning $\mathsf{Col}(R)$. Testing its 1,600 columns for membership in that public subspace therefore recovers the support and a public map U satisfying $H^\star U = R$. From one ordinary signature on h_0, an attacker transfers the syndrome equation to any fresh target h_1. A 25-bit affine correction leaves only the verifier's exact-weight condition; reflected Gray-code search finds a weight-343 representative.

The primary full-parameter run recovered all 125 columns and forged its source-frozen unsigned target after 3,657 candidates in 0.627 seconds of internal attack time, using one worker and 19,456 KiB peak resident memory. An independently written adversarial implementation forged eight additional frozen targets on the same retained key, and an exhaustive 2^{25} sweep for one target found 1,337 accepted exact-weight representatives. A fresh-context implementation then succeeded on five of five wholly fresh keys with one ordinary query per key; the slowest attack took 13.376 seconds and all controls rejected. A final scheme-specific recheck located no previous public attack against the exact target. No author code or test vectors were located; all experiments use an independently documented mathematical reconstruction of the paper. The technical, practical, adversarial, independent-reproduction, final-history, and independent-publication-review gates have all passed. The result is classified as `VERIFIED_ATTACK`.

1 Introduction

The Full-Distance Syndrome Decoding (FDSD) signature proposal presents three parameter sets intended for 128-, 192-, and 256-bit security [1]. Its verification equation resembles a random syndrome-decoding instance: the signer returns a fixed-weight binary vector, and the verifier checks its syndrome against a message-dependent public target. The Level-I row is

$$(n, k, L, \epsilon, \theta, d) = (1600, 400, 125, 5, 25, 343).$$

The target argues that a forger must solve a full-distance syndrome-decoding problem and uses the Syndrome Decoding Estimator as an indicative parameter benchmark [2].

The public matrix is not an unstructured decoding instance. Key generation preserves a 125-coordinate subset on which $H^\star$ has exactly the public column space of R. This exposes both the hidden support and an equivalent public linear transformation. A queried signature can then be moved to a new message syndrome. The only nonlinear obstacle is the prescribed weight, and the verifier gives 25 freely adjustable bits with which to restore it.

The resulting attack has the following scope:

- it targets the complete advertised Level-I row;
- it uses one ordinary signature on a message distinct from the target;
- it reads only the public matrices and that one signature;
- it outputs a weight-343 pair accepted by the ordinary verifier on a fresh attacker-fixed message; and
- it exploits the specified key structure and verification equation, rather than a parser, fault, weak random generator, or side channel.

Technique attribution. We make no novelty claim for Gaussian elimination, public subspace-membership tests, affine signature transformation, or Gray-code enumeration. The contribution is the concrete observation and full-parameter demonstration that FDSD's structured public key exposes the transformation needed to apply those standard tools and bypass the advertised decoding problem.

2 Target, reconstruction, and claim boundary

2.1 Pinned target

The target is Abdelhaliem Babiker's *A Signature Scheme from Full-Distance Syndrome Decoding*, IACR ePrint 2023/1956 [1]. The audit uses the sole object observed with archive datestamp 24 December 2023. Its 26-page PDF has SHA-256 digest

`394b7a3e4f8d935cfabffd90e17b4c8cdd5708267a14ede67b39c49ab9ead822.`

The extracted text has SHA-256 digest

`8763cd064947c50af81e487a9f7abf294b86893d2596cdff94bb199db33e460a.`

No author implementation, source repository, byte format, or test vector was located. The paper is therefore authoritative. The audit built an independent paper-level adapter and froze the following otherwise unstated conventions: the balanced group weights are eighteen 14s followed by seven 13s; Hash_{125} is the first 125 SHAKE256 output bits in MSB-first byte order; and mathematical vector coordinates are serialized MSB-first with zero padding. The attack depends on the paper's algebra, not on these particular byte choices: it accepts the same h that the verifier receives and operates entirely over $\mathbb{F}_2$ afterward.

The faithful adapter checked all 125 basic and 800 advanced key identities. On one fresh key, 4,096 of 4,096 honest signatures had exact weight 343 and were accepted; changed-message, changed-σ, changed-ζ, and changed-public-key controls rejected. The public attack process received only $H^\star$, R, and one exported ordinary signature.

2.2 Level-I public relation

All vectors and matrices below are over $\mathbb{F}_2$. Let

$$H^\star \in \mathbb{F}_2^{1200\times 1600}, \qquad R \in \mathbb{F}_2^{1200\times 125}.$$

For a message m, the reconstructed adapter computes $h = \mathsf{Hash}_{125}(m)$. The ordinary verifier accepts $(\sigma, \zeta) \in \mathbb{F}_2^{1600} \times \mathbb{F}_2^{25}$ precisely when

$$\mathsf{wt}(\sigma) = 343, \qquad H^\star\sigma = Rh + R_{[25]}\zeta, \tag{1}$$

where $R_{[25]}$ denotes the first 25 columns of R.

The claim concerns this mathematical KeyGen/Sign/Verify relation. It does not assert conformance to absent author software or an absent external signature encoding. It does not claim recovery of the nominal private key.

2.3 Attack-history status

We located no previous public attack against Abdelhaliem Babiker's exact ePrint 2023/1956 scheme in the searches completed on 14 October 2026. The search covered exact and normalized titles, identifier, author, aliases, citation indexes, public code and issue surfaces, related work, the distinct 2022/1698 predecessor, and the specific public-column recovery route. This is a bounded negative result: it cannot exclude private, deleted, or unindexed work. OpenAlex did not index the target; Semantic Scholar returned HTTP 429; unauthenticated GitHub code search returned HTTP 401; and the official all-versions endpoint returned HTTP 403.

The official landing page still lists receipt on 24 December 2023 and approval on 25 December 2023. A current mirror PDF downloaded on 14 October 2026 was byte-for-byte identical to the frozen target, with the SHA-256 digest given above; we located no later target revision. The sealed final-history handoff has SHA-256

`623f30f514d8095d558b27475dd6fe8cc122b41f84f455f02bbc79f5707d150c,`

and the sealed prior-art manifest has SHA-256

`5ffc63c382d18d2d3437092c8a0f637b256d1269f8f3120455334cc5947f0a25.`

3 Public recovery of the hidden signing map

The basic construction defines a 125-coordinate set T_e and a matrix U such that

$$HU = R. \tag{2}$$

The restriction U_{T_e} is a row/column permutation of the nonsingular middle block C_1, so it is invertible. Consequently,

$$H_{[:,T_e]}U_{T_e} = R, \qquad \mathsf{Col}(H_{[:,T_e]}) = \mathsf{Col}(R). \tag{3}$$

Advanced key generation changes columns indexed outside T_e and explicitly retains $H^\star_{[:,j]} = H_{[:,j]}$ for $j \in T_e$. It follows that every one of these 125 public columns lies in $\mathsf{Col}(R)$. The attack computes a basis for R, reduces all 1,600 columns of $H^\star$ against it, and selects the members. On the retained key it obtains exactly 125 members of rank 125; all other 1,475 columns lie outside the subspace. For random-looking outside columns, accidental membership in a 125-dimensional subspace of $\mathbb{F}_2^{1200}$ has probability 2^{-1075} per column. The implementation nevertheless checks the exact count and rank and aborts on ambiguity.

Let J be the recovered coordinate set. Solving

$$H^\star_{[:,J]}x_j = R_j$$

for every $0 \le j < 125$, and embedding each x_j into the 1,600 coordinates, gives public columns u_j with

$$H^\star u_j = R_j. \tag{4}$$

Writing $U = (u_0, \ldots, u_{124})$ yields the public equivalent transformation $H^\star U = R$. This recovers only the capability required to translate syndromes; it need not reconstruct the nominal secret-key representation.

4 One-query fresh-message forgery

Suppose an ordinary signing query on a non-target message returns (σ_0, ζ_0) for hash h_0. Then

$$H^\star\sigma_0 = Rh_0 + R_{[25]}\zeta_0. \tag{5}$$

Freeze a fresh attacker-chosen target message with hash h_1. Split each 125-bit hash into a 25-bit head and a 100-bit tail. For arbitrary $a \in \mathbb{F}_2^{25}$ define

$$\delta_{[25]} = a, \tag{6}$$

$$\delta_{[25,125)} = h_{0,[25,125)} + h_{1,[25,125)}, \tag{7}$$

$$\sigma_1(a) = \sigma_0 + U\delta, \tag{8}$$

$$\zeta_1(a) = \zeta_0 + h_{0,[25]} + h_{1,[25]} + a. \tag{9}$$

Theorem 1. *For every $a \in \mathbb{F}_2^{25}$, the pair in Equation (9) satisfies the syndrome equality in Equation (1) for target hash h_1.*

Proof. Using $H^\star U = R$,

$$H^\star \sigma_1 = Rh_0 + R_{[25]}\zeta_0 + R\delta.$$

The two copies of $h_{0,[25,125)}$ cancel, while the head contribution is $R_{[25]}a$. Regrouping the remaining terms gives

$$H^\star \sigma_1 = Rh_1 + R_{[25]}(\zeta_0 + h_{0,[25]} + h_{1,[25]} + a) = Rh_1 + R_{[25]}\zeta_1.$$

□

Thus the public equation holds identically and the attacker must only find an a for which $\mathsf{wt}(\sigma_1(a)) = 343$. The implementation walks the at most $2^{25} = 33{,}554{,}432$ values in reflected Gray-code order. Consecutive candidates differ by one recovered u_j, so each update is one XOR and one weight test. No syndrome decoding is performed.

5 Experimental evidence

5.1 Primary retained key

The sealed primary attack fixed its target before reading any public attack input. It then opened only four files: public metadata, $H^\star$, R, and exactly the first 609-byte JSONL signature record. An interpreter audit hook enforced that allowlist. No private key, seed, coordinate labels, validation summary, second signature, or network input was available.

On the fresh retained Level-I key, R had rank 125; exactly 125 columns of $H^\star$ lay in $\mathsf{Col}(R)$ and had rank 125; and all 125 recovered identities $H^\star u_j = R_j$ passed. The first exact-weight target representative appeared after 3,657 candidates. The forged vector had weight 343 and the ordinary verifier accepted. Reusing the queried signature on the target, changing the target message, changing σ, and changing ζ all rejected.

The attack reported 0.0536 seconds for public recovery, 0.00244 seconds for the correction search, and 0.627 seconds total internal time. GNU `time` measured 1.70 seconds wall time and 19,456 KiB peak RSS on one worker. A deterministic replay matched every non-timing result field.

5.2 Adversarial retained-key review

A separate AI reviewer re-derived and reimplemented the attack with the opposite pivot convention and independent row- and column-oriented verifiers. Using the same sole query, it accepted forgeries for eight source-frozen, previously unsigned targets. First solutions appeared after 6,351 to 88,507 candidates. Changed-message, unchanged-query, one-bit-ζ, and weight-preserving two-coordinate σ controls rejected for all eight.

For one target, the review exhausted all 2^{25} corrections and found 1,337 exact-weight representatives, all satisfying the public equation. The random affine-space heuristic predicts 1,339.56 for the retained query's 41 selected support coordinates. The full sweep and eight-target review took 23.70 seconds wall time and 19,456 KiB peak RSS on one worker.

5.3 Independent cross-key reproduction

A fresh-context AI agent independently implemented the attack without opening the primary attack lane before its package was sealed. It generated five wholly fresh Level-I keys from separate `os.urandom(32)` roots. For each key it fixed a distinct target before the sole ordinary signing query and gave a separate attack process only the public key and that one transcript. All five attacks succeeded, with zero target-message queries.

Every key had rank 125 for R, the recovered $H^\star$ restriction, and the recovered restriction of U; every key exposed exactly 125 public columns in $\mathsf{Col}(R)$. Candidate counts were 303, 1,014, 3,865, 199,806, and 19,856,417 (minimum/median/maximum 303/3,865/19,856,417). Per-key attack totals were 0.828834402, 13.376351877, 0.942027737, 0.835081668, and 0.849683596 seconds. Each attack process used at most

18,944 KiB RSS. The sequential wrapper, including five fresh key generations, validation, controls, and sealing, took 41.85 seconds wall time and 22,016 KiB peak RSS on one worker.

For every key the ordinary verifier accepted the weight-343 forgery and the original query, while changed-message, changed-σ, changed-ζ, wrong-key, and unchanged-query-on-target controls rejected. The sealed handoff has SHA-256

`2c7e621eadd38b5cc1386a64d2398204ac6a7a2afa14f9f7b7933bf3146b0ced,`

and the independent package manifest has SHA-256

`e6b391d30065fc8958f58514897814caef9a3658e66eba7934834f674d6f130e.`

The public reproducer has also passed an engineering smoke test: it generated a new full Level-I key without persisting the seed or secret, exported one ordinary signature, ran the isolated public forger, accepted the fresh-message forgery under ordinary Verify, and rejected all controls. This validates package operation independently of the sealed research artifacts.

6 Complexity and applicability

Public recovery performs elimination on 125 public 1,200-bit columns, reduces 1,600 public columns, and solves 125 public systems. The final enumeration is bounded by 2^{25} candidates and stores only the public matrices, a basis, 125 recovered columns, and current vectors. The retained runs used about 19 MiB peak RSS and one worker.

The syndrome transformation is deterministic once a queried signature and target are fixed. Search success depends on the 25-dimensional affine space containing a representative with the query's required support weight. The retained key exhibited many solutions and the five independent keys all yielded a solution, with first hits spanning 303 to 19,856,417 candidates. These six keys provide empirical evidence rather than a population bound. The attack detects rank loss or extra subspace columns and aborts rather than silently misclassifying them.

The route attacks the design structure. It does not depend on the chosen SHAKE256 adapter beyond obtaining two distinct 125-bit hashes, and it uses no malformed encoding, implementation error, concurrency, fault, leakage, or secret state. It presently demonstrates the advertised Level-I row; no claim is made here for the Level-III or Level-V rows.

7 Reproducibility

The public package uses Python 3.11 or later and only the standard library. Running `./reproducer/reproduce.sh` performs these steps in separate processes:

1. generate a fresh complete Level-I key from `os.urandom(32)`, sign one non-target message, and persist only public inputs;
2. freeze the target in the public forger, recover the 125-column map, search the 25-bit correction space, and write the forgery; and
3. load only the public key, query, and result into the ordinary verifier, then check freshness and four negative controls.

The successful run ends with the explicit banner `ATTACK REPRODUCTION: PASS`. Common numerical-library thread variables are pinned to one even though no numerical package is used. The generator never writes its seed, secret key, or coordinate labels. The public package is covered by a SHA-256 manifest.

8 Limitations and status

The evidence establishes a practical, full-parameter, design-level fresh-message forgery on the retained Level-I key, an eight-target adversarial review, and a five-of-five fresh-key independent reproduction. The final exact-target history recheck and independent Publication Reviewer gate also passed. All required gates passed, and the result is classified as `VERIFIED_ATTACK`. No independent human verification is required by the project Charter. The paper makes no universal all-key success-probability claim, no claim

beyond Level I, no claim of nominal secret-key recovery, and no claim that the general linear-algebra techniques are new.

9 Conclusion

FDSD's advanced public key preserves a hidden 125-column basis for the public space $\mathsf{Col}(R)$. Public subspace membership exposes that basis and reconstructs a transformation U satisfying $H^{\star}U = R$. One ordinary signature can then be translated to any fresh target syndrome, with 25 free bits left to restore the verifier's exact weight. On the retained full Level-I key the attack took well under a second internally, a separate adversarial implementation forged eight further targets, and an independent fresh-context implementation succeeded on five of five new keys. The advertised full-distance decoding problem is bypassed by a short public detour through the key's preserved column space.

Acknowledgements

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References

[1] A. Babiker. A Signature Scheme from Full-Distance Syndrome Decoding. IACR Cryptology ePrint Archive, Report 2023/1956, 2023. https://eprint.iacr.org/2023/1956.

[2] A. Esser and E. Bellini. Syndrome Decoding Estimator. In *Public-Key Cryptography—PKC 2022*, LNCS 13177, pp. 112–141. Springer, 2022. https://doi.org/10.1007/978-3-030-97131-1_5.